

Homework 1

Farid Haddad Linear Algebra Sep 17, 2026 GRADE: 35/40 (Prof. R. Castellano)

Problem 1

$$(a) \ 2u = [4, -2, 6]^T, \text{ so } 2u - v = [4, -2, 6]^T - [1, 4, -2]^T = [3, -6, 8]^T$$

$$\text{Dot that with } w: (3)(-4) + (-6)(2) + (8)(6) = -12 - 12 + 48 = 24$$

$$(2u - v) \cdot w = 24$$

[PROF: ✓]

$$(b) \ (1/2)w = [-2, 1, 3]^T, \text{ so } v + (1/2)w = [1 - 2, 4 + 1, -2 + 3]^T = [-1, 5, 1]^T$$

$$||v + (1/2)w|| = \sqrt{(-1)^2 + 5^2 + 1^2} = \sqrt{1 + 25 + 1} = \sqrt{27} = 3\sqrt{3}$$

$$3\sqrt{3} \approx 5.20$$

[PROF: ✓]

$$(c) \ \text{I need the dot product and both lengths, since } \cos \theta = (u \cdot w) / (||u|| ||w||).$$

$$u \cdot w = (2)(-4) + (-1)(2) + (3)(6) = -8 - 2 + 18 = 12$$

$$||u|| = \sqrt{4 + 1 + 9} = \sqrt{14}, \quad ||w|| = \sqrt{16 + 4 + 36} = \sqrt{56} = 2\sqrt{14}$$

$$||u|| ||w|| = \sqrt{14} \cdot 2\sqrt{14} = 2(14) = 28$$

$$\cos \theta = 12/28 = 3/7, \text{ so } \theta = \arccos(3/7)$$

$$\theta \approx 64.62^\circ$$

[PROF: -1 E8 $u \cdot w$ is 8, not 12: $(2)(-4) + (-1)(2) + (3)(6) = -8 - 2 + 18 = 8$. The slip is carried consistently to $\cos \theta = 3/7$ and $\theta \approx 64.62^\circ$, so it is deducted once.]

[PROF: Problem 1 4/5]

Problem 2

(a) AB is undefined. A is 3×3 so it has 3 columns, B is 2×3 so it has 2 rows, $3 \neq 2$.

BA is fine: $(2 \times 3)(3 \times 3) = 2 \times 3$.

row 1: $[1(2) + 0(1) + (-2)(0), 1(-1) + 0(3) + (-2)(4), 1(0) + 0(-2) + (-2)(1)] = [2, -9, -2]$

row 2: $[3(2) + 1(1) + 1(0), 3(-1) + 1(3) + 1(4), 3(0) + 1(-2) + 1(1)] = [7, 2, -1]$

$BA = \begin{bmatrix} 2 & -9 & -2 \\ 7 & 2 & -1 \end{bmatrix}$

$\begin{bmatrix} 7 & 2 & -1 \end{bmatrix}$

[PROF: -0.5 E8 BA entry (2,2) is $3(-1) + 1(3) + 1(4) = 4$, not 2.]

(b) A is 3×3 , v is 3×1 , so Av is 3×1 .

$Av = [2(1) + (-1)(-2) + 0(3), 1(1) + 3(-2) + (-2)(3), 0(1) + 4(-2) + 1(3)]^T = [4, -11, -5]^T$

vA is undefined: v has 1 column, A has 3 rows, $1 \neq 3$.

[PROF: ✓]

(c) Aw is undefined: A has 3 columns, w is 1×3 so it has 1 row, $3 \neq 1$.

wA is $(1 \times 3)(3 \times 3) = 1 \times 3$.

$wA = [2(2) + 0(1) + (-1)(0), 2(-1) + 0(3) + (-1)(4), 2(0) + 0(-2) + (-1)(1)] = [4, -6, 1]$

[PROF: -0.5 E8 wA third entry is $2(0) + 0(-2) + (-1)(1) = -1$, not 1.]

(d) vw is $(3 \times 1)(1 \times 3) = 3 \times 3$.

$$wv = \begin{bmatrix} 1(2) & 1(0) & 1(-1) \end{bmatrix} \begin{bmatrix} 2 & 0 & -1 \end{bmatrix}$$

$$\begin{bmatrix} -2(2) & -2(0) & -2(-1) \end{bmatrix} = \begin{bmatrix} -4 & 0 & 2 \end{bmatrix}$$

$$\begin{bmatrix} 3(2) & 3(0) & 3(-1) \end{bmatrix} \begin{bmatrix} 6 & 0 & -3 \end{bmatrix}$$

wv is $(1 \times 3)(3 \times 1) = 1 \times 1$, just the dot product: $wv = 2(1) + 0(-2) + (-1)(3) = -1$

[PROF: ✓]

(e) $A^2 = AA$, both 3×3 .

row 1: $\begin{bmatrix} 2(2) + (-1)(1) + 0(0), & 2(-1) + (-1)(3) + 0(4), & 2(0) + (-1)(-2) + 0(1) \end{bmatrix} = \begin{bmatrix} 3, -5, 2 \end{bmatrix}$

row 2: $\begin{bmatrix} 1(2) + 3(1) + (-2)(0), & 1(-1) + 3(3) + (-2)(4), & 1(0) + 3(-2) + (-2)(1) \end{bmatrix} = \begin{bmatrix} 5, 0, -8 \end{bmatrix}$

row 3: $\begin{bmatrix} 0(2) + 4(1) + 1(0), & 0(-1) + 4(3) + 1(4), & 0(0) + 4(-2) + 1(1) \end{bmatrix} = \begin{bmatrix} 4, 12, -7 \end{bmatrix}$

$$A^2 = \begin{bmatrix} 3 & -5 & 2 \\ 5 & 0 & -8 \\ 4 & 12 & -7 \end{bmatrix}$$

$$\begin{bmatrix} 5 & 0 & -8 \end{bmatrix}$$

$$\begin{bmatrix} 4 & 12 & -7 \end{bmatrix}$$

[PROF: -0.5 E8 A^2 entry (3,2) is $0(-1) + 4(3) + 1(4) = 16$, not 12.]

(f) $B^2 = BB$ is undefined. B is 2×3 , not square: the first B has 3 columns, the second has 2 rows, $3 \neq 2$.

[PROF: ✓]

[PROF: Problem 2 4.5/6]

Problem 3

(a) Augmented matrix and row reduce:

$$\begin{bmatrix} 1 & 2 & -1 & | & -1 \end{bmatrix}$$

$$\begin{bmatrix} 2 & 3 & 1 & | & 2 \end{bmatrix}$$

$$\begin{bmatrix} -1 & 1 & 2 & | & -1 \end{bmatrix}$$

$$R_2 \rightarrow R_2 - 2R_1 \quad \begin{bmatrix} 1 & 2 & -1 & | & -1 \end{bmatrix}$$

$$R_3 \rightarrow R_3 + R_1 \quad \begin{bmatrix} 0 & -1 & 3 & | & 4 \end{bmatrix}$$

$$\begin{bmatrix} 0 & 3 & 1 & | & 0 \end{bmatrix}$$

$$R_2 \rightarrow -R_2 \quad \begin{bmatrix} 1 & 2 & -1 & | & -1 \end{bmatrix}$$

$$R_3 \rightarrow R_3 - 3R_2 \quad \begin{bmatrix} 0 & 1 & -3 & | & -4 \end{bmatrix}$$

$$\begin{bmatrix} 0 & 0 & 10 & | & 12 \end{bmatrix}$$

$$R_3 \rightarrow (1/10)R_3 \quad \begin{bmatrix} 1 & 2 & -1 & | & -1 \end{bmatrix}$$

$$\begin{bmatrix} 0 & 1 & -3 & | & -4 \end{bmatrix}$$

$$\begin{bmatrix} 0 & 0 & 1 & | & 6/5 \end{bmatrix}$$

Three pivots, so one solution. Back substituting, $x_3 = 6/5$, then

$$x_2 = -4 + 3(6/5) = -4 + 18/5 = -2/5$$

$$x_1 = -1 - 2(-2/5) + 6/5 = -1 + 4/5 + 6/5 = 1$$

$$x = [1, -2/5, 6/5]^T$$

[PROF: -1 E8 R3 $\rightarrow$ R3 + R1 gives the constant $-1 + (-1) = -2$, not 0. Carried consistently to $x = [1, -2/5, 6/5]^T$; the correct solution is $[2, -1, 1]^T$.]

(b) $\begin{bmatrix} 1 & 2 & -1 & | & 4 \end{bmatrix}$

$$\begin{bmatrix} 2 & 5 & 1 & | & 6 \end{bmatrix}$$

$$\begin{bmatrix} 3 & 7 & 0 & | & 10 \end{bmatrix}$$

$$R_2 \rightarrow R_2 - 2R_1 \quad \begin{bmatrix} 1 & 2 & -1 & | & 4 \end{bmatrix}$$

$$R_3 \rightarrow R_3 - 3R_1 \quad \begin{bmatrix} 0 & 1 & 3 & | & -2 \end{bmatrix}$$

$$\begin{bmatrix} 0 & 1 & 3 & | & -2 \end{bmatrix}$$

$$R_3 \rightarrow R_3 - R_2 \quad \begin{bmatrix} 1 & 0 & -7 & | & 8 \end{bmatrix}$$

$$R_1 \rightarrow R_1 - 2R_2 \quad [0 \ 1 \ 3 \mid -2]$$

$$[0 \ 0 \ 0 \mid 0]$$

Column 3 has no pivot, so z is free. Let $z = t$.

Row 2: $y = -2 - 3t$. Row 1: $x = 8 + 7t$.

Infinitely many solutions, one for each value of t .

$$[x, y, z]^T = [8, -2, 0]^T + t[7, -3, 1]^T, t \in \mathbb{R}$$

[PROF: ✓]

[PROF: Problem 3 5/6]

Problem 4

(a) Let x_1, x_2, x_3 be the masses, in kg, of Alloy 1, Alloy 2, Alloy 3 that get used. The 100 kg product holds 36 kg copper, 44 kg zinc, 20 kg nickel, and nothing is lost,

so for each metal the mass going in equals the mass coming out.

$$\text{copper: } 0.5x_1 + 0.2x_2 + 0.4x_3 = 36$$

$$\text{zinc: } 0.3x_1 + 0.7x_2 + 0.2x_3 = 44$$

$$\text{nickel: } 0.2x_1 + 0.1x_2 + 0.4x_3 = 20$$

[PROF: ✓]

(b) Multiply each row by 10 to clear the decimals:

$$[5 \ 2 \ 4 \mid 360]$$

$$[3 \ 7 \ 2 \mid 400]$$

$$[2 \ 1 \ 4 \mid 200]$$

[PROF: -0.5 E8 The zinc row constant is $10(44) = 440$, not 400.]

$$(c) \quad [5 \ 2 \ 4 \mid 360]$$

$$[3 \ 7 \ 2 \mid 440]$$

$$[2 \ 1 \ 4 \mid 200]$$

$$R1 \rightarrow R1 - 2R3 \quad [1 \ 0 \ -4 \mid -40]$$

$$[3 \ 7 \ 2 \mid 440]$$

$$[2 \ 1 \ 4 \mid 200]$$

$$R2 \rightarrow R2 - 3R1 \quad [1 \ 0 \ -4 \mid -40]$$

$$R3 \rightarrow R3 - 2R1 \quad [0 \ 7 \ 14 \mid 520]$$

$$[0 \ 1 \ 12 \mid 280]$$

$$R2 \rightarrow (1/7)R2 \quad [1 \ 0 \ -4 \mid -40]$$

$$R3 \rightarrow R3 - R2 \quad [0 \ 1 \ 2 \mid 520/7]$$

$$[0 \ 0 \ 10 \mid 1440/7]$$

$$R3 \rightarrow (1/10)R3 \quad [1 \ 0 \ -4 \mid -40]$$

$$[0 \ 1 \ 2 \mid 520/7]$$

$$[0 \ 0 \ 1 \mid 144/7]$$

$$x_3 = 144/7 \approx 20.57, \quad x_2 = 520/7 - 2(144/7) = 232/7 \approx 33.14$$

$$x_1 = -40 + 4(144/7) = -40 + 576/7 = 296/7 \approx 42.29$$

Use about 42.29 kg of Alloy 1, 33.14 kg of Alloy 2, and 20.57 kg of Alloy 3.

[PROF: -1 E8 From the matrix as written, $R2 \rightarrow R2 - 3R1$ gives $440 - 3(-40) = 560$, not 520.

Carried consistently to 42.29, 33.14, 20.57 kg; the correct masses are 40, 40, 20 kg.]

[PROF: Problem 4 6.5/8]

Problem 5

(a) $\begin{bmatrix} 1 & 2 & 1 \end{bmatrix}$

$$\begin{bmatrix} 2 & 4 & 3 \end{bmatrix}$$

$$\begin{bmatrix} 3 & 6 & 4 \end{bmatrix}$$

$$R_2 \rightarrow R_2 - 2R_1 \quad \begin{bmatrix} 1 & 2 & 1 \end{bmatrix}$$

$$R_3 \rightarrow R_3 - 3R_1 \quad \begin{bmatrix} 0 & 0 & 1 \end{bmatrix}$$

$$\begin{bmatrix} 0 & 0 & 1 \end{bmatrix}$$

$$R_3 \rightarrow R_3 - R_2 \quad \begin{bmatrix} 1 & 2 & 0 \end{bmatrix}$$

$$R_1 \rightarrow R_1 - R_2 \quad \begin{bmatrix} 0 & 0 & 1 \end{bmatrix}$$

$$\begin{bmatrix} 0 & 0 & 0 \end{bmatrix}$$

$$\text{RREF}(A) = \begin{bmatrix} 1 & 2 & 0 \end{bmatrix}$$

$$\begin{bmatrix} 0 & 0 & 1 \end{bmatrix}$$

$$\begin{bmatrix} 0 & 0 & 0 \end{bmatrix}$$

[PROF: ✓]

(b) $\text{rank}(A) = 2$, because the RREF has pivots in columns 1 and 3, so two pivots (two nonzero rows).

[PROF: ✓]

(c) The same operations on $[A \mid b]$ leave the bottom row as $[0 \ 0 \ 0 \mid b_3 - b_1 - b_2]$.
0 solutions: possible. If $b_3 \neq b_1 + b_2$, for example $b = [0, 0, 1]^T$, that row says $0 = 1$, so the system is inconsistent.

Exactly 1 solution: impossible. Column 2 never holds a pivot, so x_2 is free whenever the system is consistent. $\text{rank}(A) = 2 < 3$ unknowns, so the solution is never unique.

Infinitely many: possible. If $b_3 = b_1 + b_2$, say $b = [1, 3, 4]^T$ or $b = 0$, the system is consistent and the free variable x_2 gives infinitely many solutions.

[PROF: ✓]

[PROF: Problem 5 7/7]

Problem 6

(a) vectors as columns, then row reduce:

$$\begin{bmatrix} 1 & 2 & 0 \end{bmatrix}$$

$$\begin{bmatrix} 0 & 1 & 1 \end{bmatrix}$$

$$\begin{bmatrix} 2 & 3 & -1 \end{bmatrix}$$

$$R_3 \rightarrow R_3 - 2R_1 \quad \begin{bmatrix} 1 & 2 & 0 \end{bmatrix}$$

$$\begin{bmatrix} 0 & 1 & 1 \end{bmatrix}$$

$$\begin{bmatrix} 0 & -1 & -1 \end{bmatrix}$$

$$R_3 \rightarrow R_3 + R_2 \quad \begin{bmatrix} 1 & 2 & 0 \end{bmatrix}$$

$$\begin{bmatrix} 0 & 1 & 1 \end{bmatrix}$$

$$\begin{bmatrix} 0 & 0 & 0 \end{bmatrix}$$

Two pivots for three vectors, so there is a free variable and the set is dependent.

The relation is $v_3 = v_2 - 2v_1$, since $[2, 1, 3]^T - [2, 0, 4]^T = [0, 1, -1]^T$, and v_1, v_2 are not multiples of each other so they still span two dimensions.

Dependent, and the span is a plane through the origin in $\mathbb{R}^3$.

[PROF: ✓]

(b) $\begin{bmatrix} 1 & 0 & 1 \end{bmatrix}$

$$\begin{bmatrix} 1 & 1 & 0 \end{bmatrix}$$

$$\begin{bmatrix} 0 & 1 & 1 \end{bmatrix}$$

$$R_2 \rightarrow R_2 - R_1 \quad \begin{bmatrix} 1 & 0 & 1 \end{bmatrix}$$

$$\begin{bmatrix} 0 & 1 & -1 \end{bmatrix}$$

$$\begin{bmatrix} 0 & 1 & 1 \end{bmatrix}$$

$$R_3 \rightarrow R_3 - R_2 \quad \begin{bmatrix} 1 & 0 & 1 \end{bmatrix}$$

$$\begin{bmatrix} 0 & 1 & -1 \end{bmatrix}$$

$$\begin{bmatrix} 0 & 0 & 2 \end{bmatrix}$$

Three pivots for three vectors, so $c_1 v_1 + c_2 v_2 + c_3 v_3 = 0$ only when $c_1 = c_2 = c_3 = 0$, which is what independent means, and 3 of them fill $\mathbb{R}^3$. Independent, and the span is all of $\mathbb{R}^3$.

[PROF: ✓]

[PROF: Problem 6 8/8]

Grader summary

Problem 1: 4/5

Problem 2: 4.5/6

Problem 3: 5/6

Problem 4: 6.5/8

Problem 5: 7/7

Problem 6: 8/8

Total: 35/40

Your method is correct throughout and every row operation is labeled in the required form. Every point you lost is arithmetic: $u \cdot w$ in 1(c), three matrix entries in Problem 2, and constants in 3(a), 4(b), and 4(c). Check results against the problem statement — the masses in 4(c) sum to 96 kg, not the 100 kg you were given.